

Spencer D. Kelly

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Education

University of California, Los Angeles (UCLA)

Master of Science, Mechanical Engineering – Robotics Specialization

Bachelor of Science, Bioengineering

Los Angeles, California

June 2026

June 2025

Undergraduate GPA: 3.72 | Graduate GPA: 3.84

Skills

Coding	MATLAB, C++, ACSPL+ (ACS Motion Control), Python, Java
Software	SolidWorks, Onshape, Simulink, COMSOL, OpenSim, OpenCV, TOPS, Processing, MuJoCo
Hardware	Soldering, Circuitry, Microcontrollers, Optical Sensors, MIG Welding, Plasma Cutting, CNC Machining
Languages	English (native), Spanish, Italian

Experience

Ultima Genomics

Fremont, California

Systems Automation Engineering Intern

June 2026 – September 2026

- Created and executed 30+ automated test procedures to validate motion, sensor, and mechanical performance of a new robotic system, ensuring readiness for future production use and release
- Developed automated test scripts on ACS precision motion controller as part of design verification testing of the lead screw backlash, move repeatability, and lifetime performance for a 3-DoF linear system
- Designed a multi-sensor tilt-sensing scheme and integrated it into an automated pick-and-place sequence combining motion control, vacuum actuation, and pressure sensing for silicon wafer chucking for 3000+ cycles
- Characterized lead screw backlash at 60 μm over 2500 cycles against a 10-100 μm vendor spec, enabling a software compensation scheme that cut residual positioning error to under 10 μm

3-Way Machining

Celina, Ohio

CNC Machining Intern

July 2025 – September 2025

- Operated 4-axis CNC mill with G-code and conversational programming for precision industrial components
- Utilized Fusion 360 (CAD/CAM) to design specialized components for steel and aluminum automotive fixtures
- Optimized CNC toolpaths to improve efficiency, achieving a 31% cycle time reduction on standard order parts
- Documented and performed quality inspections with CMMs, calipers, and micrometers to verify dimensional accuracy to .001" and compliance with GD&T specifications

Applied Medical

Rancho Santa Margarita, California

Process Engineering Intern

June 2024 – September 2024

- Installed and tested blade design upgrades on industrial thermoformers used for sterile medical packaging
- Reworked pattern recognition vision system (Cognex) to correct for infrared attenuation from labels, leading to a 187% increase in throughput and 55% reduction in downtime for sterilization pack out procedures
- Implemented a plan to reduce outsourcing of sealing tray lids, cutting back expenditures by 71% per lid unit
- Conducted equipment validation with installation qualifications (IQs) and process qualifications (PQs) to promote safety and efficiency within the manufacturing process, compliant with FDA and ISO standards

Project Experience

Autonomous Air Hockey Table Robot

Team Lead

March 2026 – June 2026

- Spearheaded the development of a 2-DoF gantry-led robotic air hockey defender on a Raspberry Pi 4B, successfully integrating computer vision, predictive control, and hardware subsystems for blocking capabilities
- Implemented HSV color-space vision pipeline in Python (OpenCV, Picamera2) with a custom calibration tool for real-time object tracking and threshold adjustment, developing into a kinematic trajectory algorithm
- Designed and tuned a decentralized task-space PID controller with EMA-filtering, integral windup clamping, and deadband logic, driving Dynamixel servos to achieve a 1.54cm RMS tracking error and 0.85s settling time

Cake Icing Robot

Electronics Lead

September 2025 – December 2025

- Implemented the electronics and control system for a 5-DoF robotic manipulator (3 rotational, 2 linear joints)
- Integrated high-accuracy positional control of Dynamixel MX-28AR servos via an ESP32 using RS-485 Protocol 2.0
- Validated motion planning and control algorithms for smooth multi-DoF trajectories using DH parameterization
- Developed custom packet-level functions to configure servo operating modes, torque profiles, and real-time behavior to meet kinematic requirements